Question			Answer	Marks	AOs	Guidance	
6			$\int_a^b \left(x^3 - 5x^2 + 7x - 3\right) \mathrm{d}x$	**M1**	**2.1**	All three brackets expanded. Allow sign errors and one coefficient error	Must be 4 term cubic
				A1	**1.1**	All correct	
			$a = 1$ and $b = 3$	**B1**	**2.2a**		
			$\mathrm{F}[x] = \frac{x^4}{4} - \frac{5x^3}{3} + \frac{7x^2}{2} - 3x$	**M1**	**1.1**	integration of their expanded brackets; condone $+c$ Allow sign errors and one coefficient error	Later M marks both dep on first M1
			F [3] – F[1] evaluated	**M1**	**1.1**	Subst seen, allow sign or num slip $\left[\frac{81}{4} - 45 + \frac{63}{2} - 9\right] - \left[\frac{1}{4} - \frac{5}{3} + \frac{7}{2} - 3\right]$	$\left[-\frac{9}{4}\right] - \left[-\frac{11}{12}\right]$
			$= -\frac{4}{3}$ so required area is $\left\lvert -\frac{4}{3} \right\rvert = \frac{4}{3}$ AG	**A1**	**2.4**	Sign change must be seen; no reliance on decimals	
				[6]			